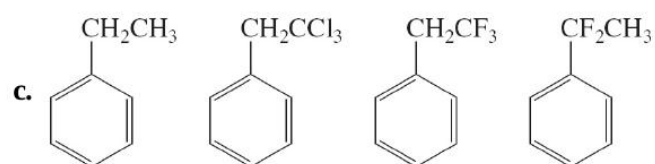
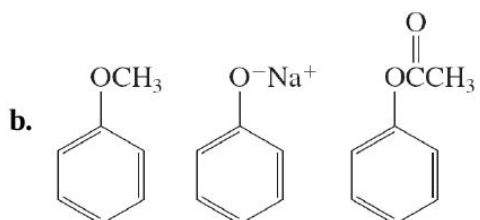
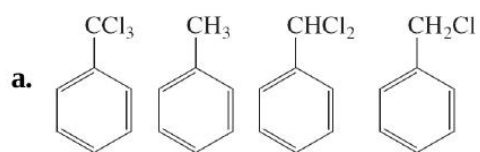
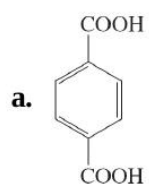


30. Rank the compounds in each of the following groups in order of decreasing reactivity toward electrophilic substitution. Explain your rankings.

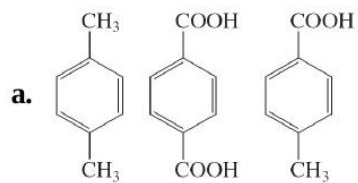


31. The rate of nitration of (chloromethyl)benzene, , is 0.71 relative to the rate of nitration of benzene (=1).⁽⁼¹⁾ The (chloromethyl)nitrobenzene product mixture that results contains 32% 32% ortho, 15.5%15.5% meta, and 52.5%52.5% para isomers. Explain.

32. Specify whether you expect the benzene rings in the following molecules to be activated or deactivated.



33. Rank the compounds in each of the following groups in order of decreasing reactivity toward electrophilic aromatic substitution. Explain your answers.



34. Halogenation of 1,3-dimethylbenzene (*m*-xylene) takes place 100 times faster than halogenation of either its 1,2- or 1,4-isomers (*o*- or *p*-xylene). Suggest an explanation.

35. Write the structure(s) of the major product(s) that you expect from each of the following electrophilic aromatic substitutions. **(a)** Nitration of methylbenzene (toluene); **(b)** sulfonation of methylbenzene (toluene); **(c)** nitration of 1,1-dimethylethylbenzene (*tert*-butylbenzene); **(d)** sulfonation of 1,1-dimethylethylbenzene (*tert*-butylbenzene).

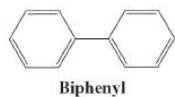
In what way does changing the substrate structures from methylbenzene (toluene) in (a) and (b) to 1,1-dimethylethylbenzene (*tert*-butylbenzene) in (c) and (d) affect the expected product distributions?

- 36.** Write the structure(s) of the major product(s) that you expect from each of the following electrophilic aromatic substitutions. **(a)** Sulfonation of methoxybenzene (anisole); **(b)** bromination of nitrobenzene; **(c)** nitration of benzoic acid; **(d)** Friedel-Crafts acetylation of chlorobenzene.

37. Draw appropriate resonance forms to explain the deactivating meta-directing character of the SO_3H group in benzenesulfonic acid.

- 38.** Do you agree with the following statement? “Strongly electron-withdrawing substituents on benzene rings are meta directing because they deactivate the meta positions less than they deactivate the ortho and para positions.” Explain your answer.

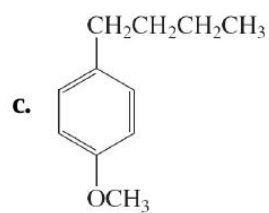
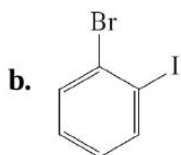
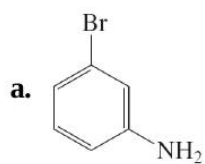
- 39.** Draw appropriate resonance forms to explain the activating ortho, para-directing character of the phenyl substituent in biphenyl (below).



40. Give the expected major product(s) of each of the following

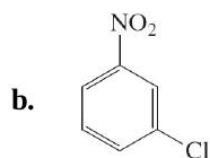
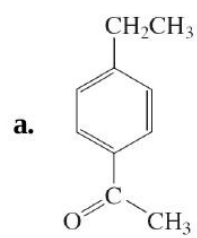
41. Reaction review. Without consulting the Reaction Road Map on [p. 749](#), suggest a combination of reagent and monosubstituted benzene that would give each of the following compounds. (**Hint:** Refer to [Table 16-2](#) for guidance. **Caution:** Directing effects arise from the group already

42. Reaction review. The preparation of the following compounds requires more than one step. As in [Problem 41](#), suggest a monosubstituted benzene as starting material and the reagents for all the steps needed to complete their synthesis.



44. **CHALLENGE** (a) When a mixture containing one mole each of the three dimethylbenzenes (*o*-, *m*-, and *p*-xylene) is treated with one mole of chlorine in the presence of a Lewis acid catalyst, one of the three hydrocarbons is monochlorinated in 100%~~100%~~ yield, whereas the other two remain completely untouched. Which isomer reacts? Explain the differences in reactivity. (b) The same experiment carried out on a mixture of the following three trimethylbenzenes gives a similar outcome. Answer the questions posed in (a) for this mixture of isomers.

45. Propose a reasonable synthesis of each of the following multiply substituted arenes from benzene.



46. (4-Methoxyphenyl)methanol (anisyl alcohol) (see below) contributes both to the flavor of licorice and to the fragrance of lavender. Propose a synthesis of this compound from methoxybenzene (anisole). (**Hint:**

47. Propose a one-step synthesis of sulfosalicylic acid ([p. 775](#)) from a suitable precursor.

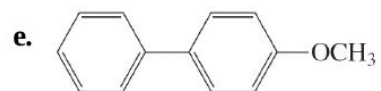
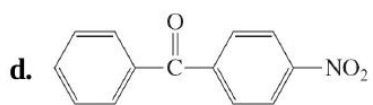
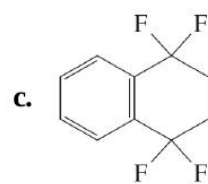
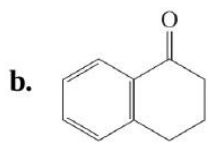
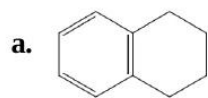
50. Catalytic hydrogenation of naphthalene over Pd-C results in rapid addition of 2 moles of H₂. Propose a structure for this product.

51. Predict the major mononitration product of each of the following disubstituted naphthalenes. **(a)** 1,3-Dimethylnaphthalene; **(b)** 1-chloro-5-methoxynaphthalene; **(c)** 1,7-dinitronaphthalene; **(d)** 1,6-

53. **CHALLENGE** Sulfonation of naphthalene at 80°C gives almost entirely 1-naphthalenesulfonic acid, whereas the same reaction at 160°C gives 2-naphthalenesulfonic acid. Propose an explanation. (**Hint:** For the underlying principle, see [Section 14-6](#).)

55. **(a)** Although methoxy is a strongly activating (and ortho, para-directing) group, the meta positions in methoxybenzene (anisole) are actually slightly deactivated toward electrophilic substitution relative to benzene. Explain. **(b)** Draw a set of potential-energy curves similar to those in [Section 16-2](#), comparing the electrophilic substitution of benzene with that of methoxybenzene (anisole) in ortho, para and meta positions.

56. Predict the result of mononitration of



- 58.** Typical conditions for nitrosation are illustrated in the following equation. Propose a detailed mechanism for this reaction.